CISCO PACKET TRACER PROJECT

Design and Configure a Secure Office Network Using VLANs

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2 Network Topology

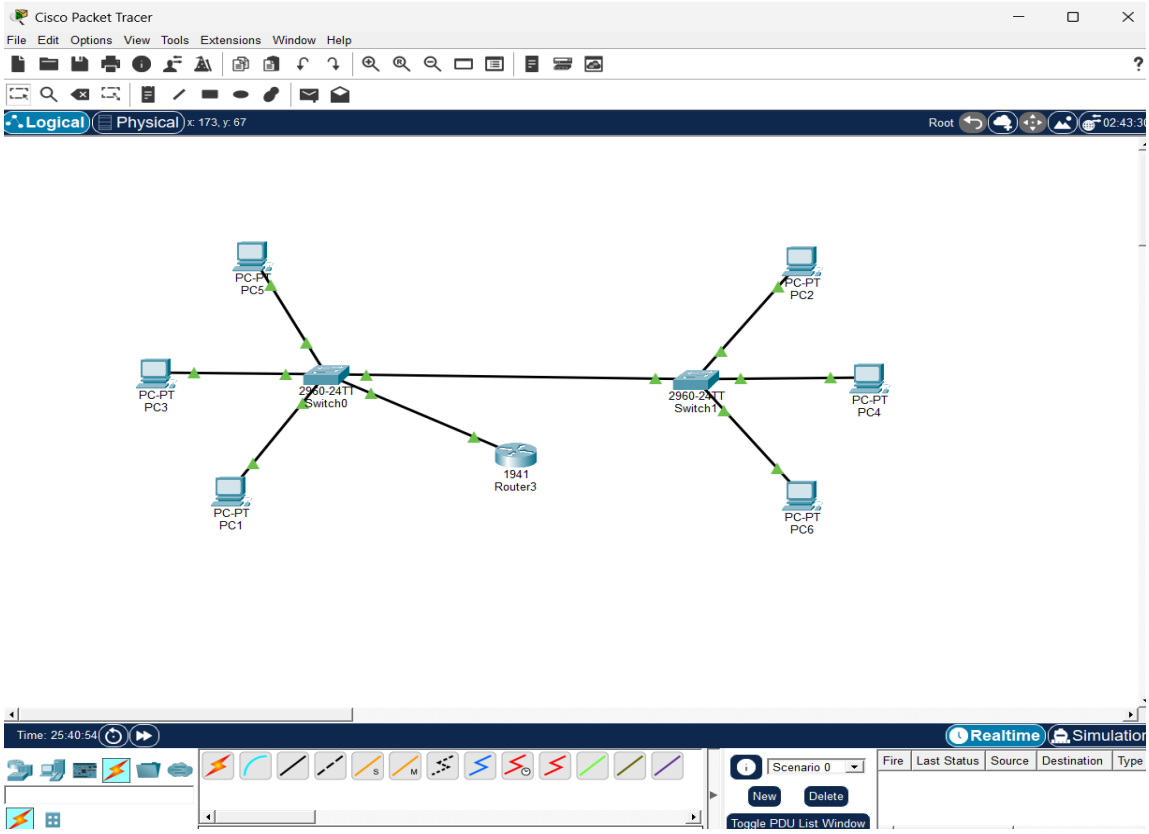


Figure 1: physical topology

3 Vlan configuration

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name IT
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name FINANCE
Switch(config-vlan)#exit
Switch(config)#! Configure HR PC Port (VLAN 10)
Switch(config)#interface fastethernet 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
Switch(config)#! Configure IT PC Port (VLAN 20)
Switch(config)#interface fastethernet 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#
Switch(config)#! Configure Finance PC Port (VLAN 30)
Switch(config)#interface fastethernet 0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
```

Figure 2: VLAN-Switch0

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#! 1. Create the VLANs
Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name IT
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name FINANCE
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#! 2. Assign the PC ports plugged into Switch1
Switch(config)#! (Adjust the port numbers if your PCs are plugged into different ports on Switch1)
Switch(config)#interface fastethernet 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fastethernet 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fastethernet 0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
```

Figure 3: VLAN-Switch1

4 Trunk configuration

```
Switch(config)#interface gigabitethernet 0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface gigabitethernet 0/2
Switch(config-if)#switchport mode trunk

Switch(config-if)#exit
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down
```

Figure 4: Trunk-Switch0

```
Switch(config)#
Switch(config)#! 3. Configure the trunk link connecting Switch1 to Switch0
Switch(config)#! (Check which port the cable between the two switches is plugged into, e.g.,
gigabitethernet 0/1)
Switch(config)#interface gigabitethernet 0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 5: Trunk-Switch1

5 Router

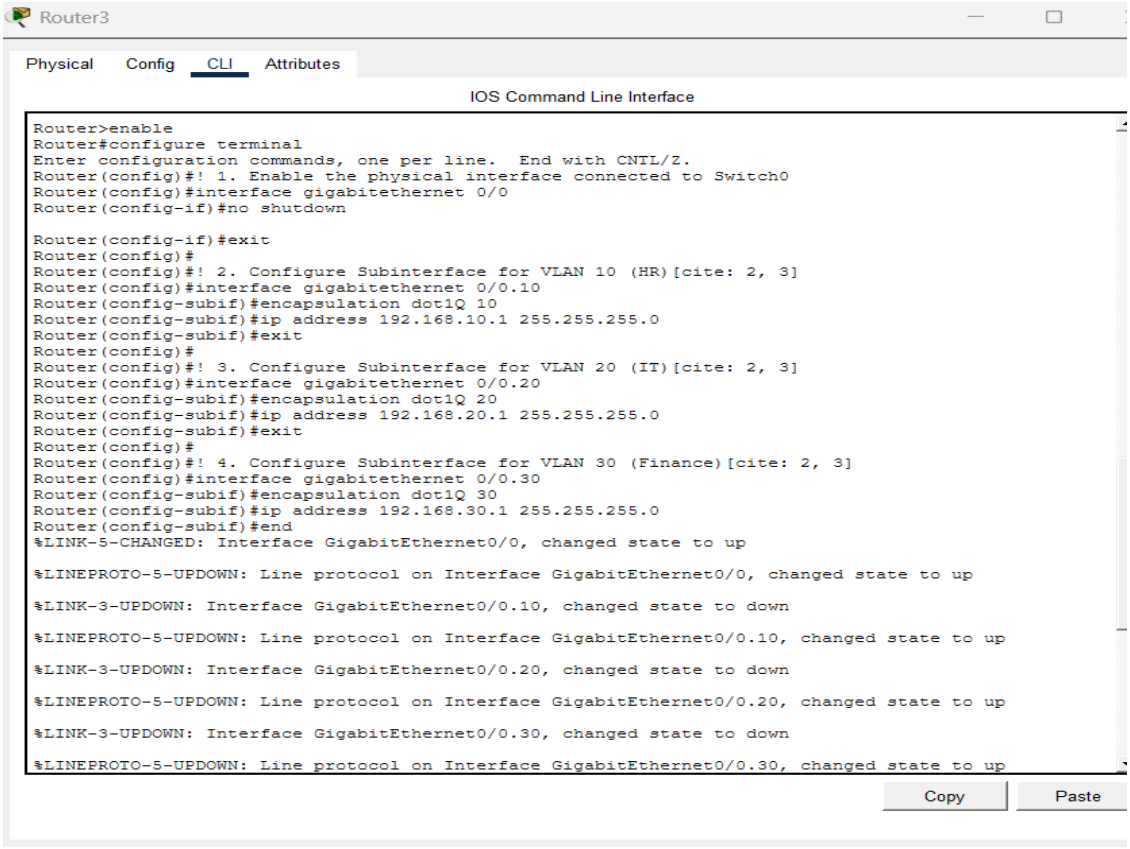


Figure 6: Router Subinterfaces

6.1 Same-Vlan

```
C:\>ping 192.168.20.11

Pinging 192.168.20.11 with 32 bytes of data:

Reply from 192.168.20.11: bytes=32 time<1ms TTL=127
Reply from 192.168.20.11: bytes=32 time=10ms TTL=127
Reply from 192.168.20.11: bytes=32 time<1ms TTL=127
Reply from 192.168.20.11: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.20.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping 192.168.30.11

Pinging 192.168.30.11 with 32 bytes of data:

Reply from 192.168.30.11: bytes=32 time<1ms TTL=127
Reply from 192.168.30.11: bytes=32 time<1ms TTL=127
Reply from 192.168.30.11: bytes=32 time=10ms TTL=127
Reply from 192.168.30.11: bytes=32 time=10ms TTL=127

Ping statistics for 192.168.30.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 5ms
```

```
C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time<1ms TTL=127
Reply from 192.168.10.11: bytes=32 time=1ms TTL=127
Reply from 192.168.10.11: bytes=32 time=11ms TTL=127
Reply from 192.168.10.11: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 3ms
```

6.2 Inter-Vlan

```
C:\>ping 192.168.10.12

Pinging 192.168.10.12 with 32 bytes of data:

Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time=1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Command Prompt X

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.12

Pinging 192.168.20.12 with 32 bytes of data:

Reply from 192.168.20.12: bytes=32 time<1ms TTL=128
Reply from 192.168.20.12: bytes=32 time=1ms TTL=128
Reply from 192.168.20.12: bytes=32 time<1ms TTL=128
Reply from 192.168.20.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

```
Pinging 192.168.30.12 with 32 bytes of data:

Reply from 192.168.30.12: bytes=32 time<1ms TTL=128
Reply from 192.168.30.12: bytes=32 time<1ms TTL=128
Reply from 192.168.30.12: bytes=32 time<1ms TTL=128
Reply from 192.168.30.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.30.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

8 Troubleshooting Report

Fault 1: IP Address Subnet Mismatch ("Destination Host Unreachable")

Description: During initial testing, a PC failed to reach other subnets and returned a "Destination host unreachable" error.

Root Cause: A typo in the PC's IPv4 configuration (typing 192.16. instead of the correct 192.168. network prefix).

Solution: Corrected the IP address octet to align with the proper VLAN subnet (192.168.), restoring valid communication with the default gateway.

Fault 2: Inter-VLAN Routing Drop

Description: Inter-VLAN pings initially timed out before configuration was fully verified.

Root Cause: Missing or incorrect 802.1Q encapsulation mapping on router subinterfaces, or access ports assigned to mismatched VLANs on the switches.

Solution: Verified that all router subinterfaces had matching encapsulation dot1Q [VLAN_ID] commands applied prior to setting their IP addresses and ensured switch access ports were correctly mapped using switchport access vlan.

9 Brief Conlcusion

This project successfully demonstrated the implementation of a multi-VLAN office network utilizing a Router-on-a-Stick architecture. By segmenting the physical network into logical Virtual Local Area Networks (HR, IT, and Finance), broadcast domains were effectively reduced, leading to optimized network performance and enhanced security. Furthermore, configuring 802.1Q trunking and router subinterfaces enabled seamless inter-departmental communication while maintaining strict, logical isolation, proving that VLANs provide a scalable and cost-effective alternative to physical routing hardware for modern enterprise networks.